Area Efficient Carry Look ahead Adder Based on Memristor Ratioed Logic

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***Abstract*— There is always a need for high-performance computing in the current technological growth and is increasing day by day. To meet this requirement normal transistors cannot match to provide the required performance requirements. Memristors are capable of a wide range of applications from signal processing, neuromorphic computing, quantum computing applications. The integration of memristor-based digital logic circuits known as memristor ratioed logic (MRL) opens new ways for exploring High-Performance Computing (HPC) architectures. In this paper, Carry Look Ahead adders (CLA) using MRL logic has been presented. By using MRL universal gate (MRLUG) and modifying architecture demonstrated to reduce the area. The designed circuit is an area efficient 4-bit CLA adder implemented in cadence virtuoso, which shows that the design requires fewer memristors which consume less power.**

Keywords—Memristor, MRL, HPC, CLA.

I. INTRODUCTION

Traditional CMOS suffers from short channel effects, fabrication effects, so people are looking for alternative technologies like memristor. The memristor is a high-speed device. Memristors are the 4th fundamental element that describes the relation between charge and magnetic flux, as described in Figure 1. It is non-volatile, passive, non-linear. It's normally a resistor with the feature of storing the memory. It is an alternative for DRAM memories. Leon Chua first described the term memristor while working with the Non-linear relationship between the fundamental quantities. Memristor is a basic device which are used in analog, digital memories & circuits. HP Labs fabricated the first memristor in 2008. It doesn't require a power source to switch ON the device. The memristor is considered like a water pipe. When there is a flow of water, it acts like a normal resistor, but when there is no flow of water, then it remembers the last known state. The diameter of pipe remains last state; thus, it shows that it acts as a memory element.

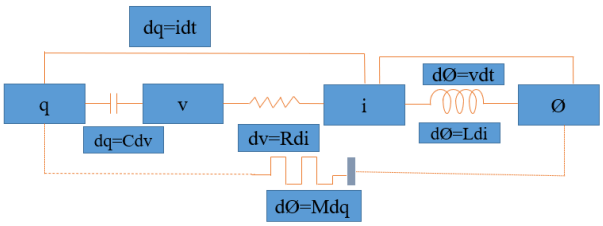


Fig. 1. Representation of memristor as fundamental element

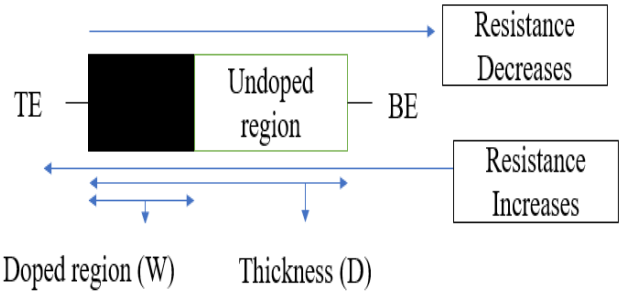


Fig. 2. Symbol of memristor

Figure 2 shows the memristor symbol, and the direction shows whether resistance decrease or increases based on the direction of input applied. When the input (Voltage/current) is given as input to one end of the memristor, it shows High Resistive State (HRS), and while the input is given to the other end, it shows Less Resistive State (LRS), as shown in Figure 3. The undoped region is TiO_2 region. The doped region (TiO_{2-x}) is variable. When input is applied, oxygen vacancies present in TiO_{2-x} are attracted by the opposite charge polarity so the carriers drift changing the resistance states.

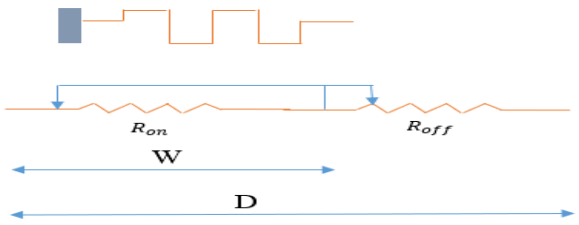


Fig. 3. Low Resistive State.

Zinc oxide (ZnO), Hafnium oxide (HfO_2), Vanadium dioxide (VO_2), tantalum oxide (Tao_x) can also be used as a Metal oxide active layer. Gold (Au), Silver (Ag), Aluminium (Al), Platinum (Pt) can be used as TE or BE electrode materials. The black shaded region is the doped region that has Positive polarity. W/D decides Noise margin. The doped region changes based on the input supply applied. The effect of temperature is very negligible. On changing the W memristance changes, so the output voltage changes. It is given as

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$$V = M(q).\ I \tag{1}$$

Where M = memristance [5]

$$M = R_{on}\frac{W}{D} + R_{off}\left(1 - \frac{W}{D}\right) \tag{2}$$

Where $\frac{W}{D} \in [0,1]$

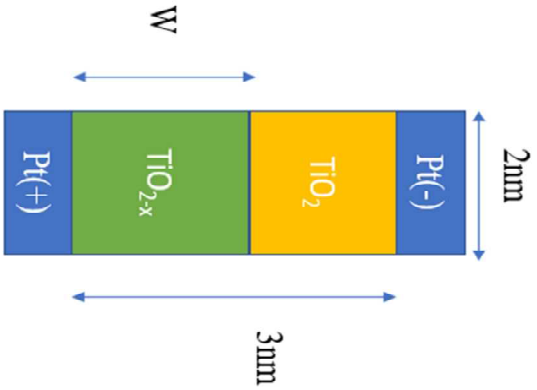


Fig. 4. Memristor design

This paper uses Platinum (Pt), Titanium dioxide (Tio_2) are used to implement the memristor, as shown in Figure 4. There are several memristor-based logical circuit design methodologies they are IMPLY, MAGIC, MRL, X-MRL. They are the same as ECL, TTL CMOS in the transistors.

IMPLY is the simple method to implement logic [1] that involves memristors and resistors. IMPLY Condition says that a true statement never implies a false statement. To design AND logic gate using IMPLY based logic, it requires 9 Memristors. Complexity in logic is the biggest disadvantage.

Memristor-aided Logic (MAGIC) relies on a transmission of inputs to give the Logic [1]. In this Logic, high resistance shows 0 logic state, whereas low resistance shows 1 logic state. The circuit is simpler and more stable. As shown in [1]. To design AND logic gate using MAGIC based Logic requires 3 Memristors. But the disadvantage is there are not compatible with the traditional CMOS.

The Memristor Ratioed Logic (MRL) Logic shows that any logic circuit can be done by using combination of memristor and CMOS logic. The logic levels are the same as in CMOS. This paper proposes Area efficient carry-lookahead adder (CLA) done using MRL method. A performance comparison between Modified CLA and existing CLAs has been conducted. Section II presents the memristor modelling. Section III is CLA adder based on MRLUG logic. Section IV shows the Area efficient CLA adder. Section V and Section VI presents the Results and Discussion, Conclusion respectively.

II. MODELLING THE MEMRISTOR

Several models like VTEAM, HP Model, Yakopcic, etc., available to model the properties of the memristor. The memristor performance entirely depends on the equations used in the model [11].

The $i - v$ relation of the model is given:

$$i(t) = \begin{cases} a_1 x(t)\sin(bv(t)), & v(t) \geq 0 \\ a_2 x(t)\sin(bv(t)), & v(t) < 0 \end{cases} \tag{3}$$

Where $i(t)$ & $v(t)$ are voltage and current across the memristor. $x \in [0,1]$ is the internal state variable (ISV) of the memristor, which satisfies:

$$\frac{dx}{dt} = g(v(t))\, f(x(t)) \tag{4}$$

Where $g(v(t))$ imposes different threshold on model given by

$$g(v(t)) = \begin{cases} A_p\left(e^{v(t)} - e^{V_p}\right), & v(t) > V_p \\ -A_n\left(e^{-v(t)} - e^{V_n}\right), & v(t) < V_n \\ 0, & -V_n \leq v(t) \leq V_p \end{cases} \tag{5}$$

When internal state variable reaches boundaries of memristor $x_p$and x_n. Biolek window, Jogelkar window, etc., have been described to make the device switch smoothly at the boundaries. The $i - v$ curve is shown in Figure 5. When the input is applied, memristance will transfer from HRS (High Resistance State) to LRS (Low Resistance State) [1]. The carrier concentration changes. Then the memristor is in SET state. When the applied voltage greater the positive threshold voltage Memristance will transfer from LRS (Low Resistance State) to HRS (High Resistance State) when the applied voltage lesser then the negative threshold voltage [1]. The memristor is in RESET state. At higher frequencies, the curve will be linear.

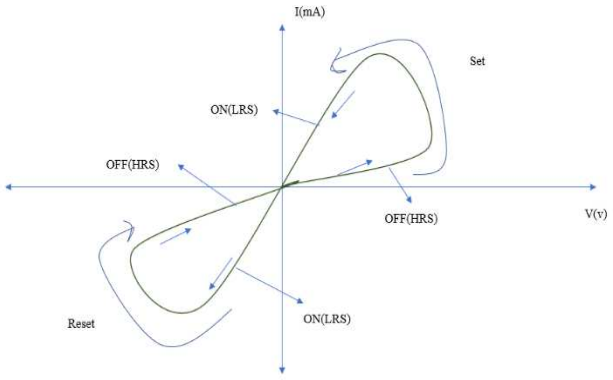


Fig. 5. $i - v$ characteristic curve of memristor.

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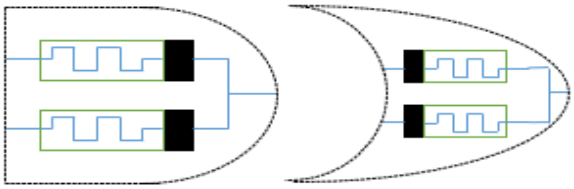


Fig. 6. Memristor logic of AND and OR logic

The process is analyzed by taking AND gate as an example shown in Figure 6. The input terminal is connected to the undoped terminal of the two memristors, and both the doped terminals are shorted. The output is taken from the shorted terminal. If both the inputs are same, then the output is floating, so the output is same as the input. When both the inputs are different, then $R_{off} >> R_{on}$. So, the output voltage is calculated by the principle of the voltage divider.

$$V_{out} = \frac{R_{on}}{R_{on}+R_{off}} * V_{dd} \quad (6)$$

AND gate, and a CMOS Inverter gives the NAND logic gate. The output logic states of MRL are voltage levels. So, it can be used with traditional logic circuits.

III. CLA ADDER BASED ON MRLUG

A 4-bit CLA based on MRL is given in [1]. The circuit is constructed by using the Universal logic gate (ULG), which is used to produce AND, XOR, OR logics at one go.

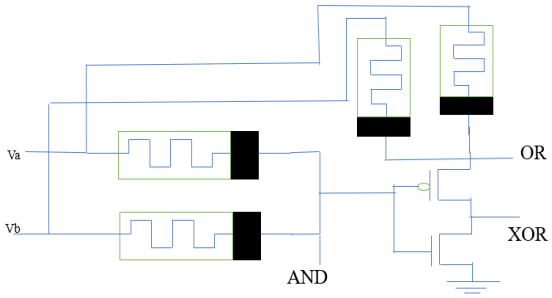


Fig. 7. ULG with MRL logic

The CLA functions are:

$$S_i = a_i \hat{} b_i \hat{} c_i \quad c_{i+1} = G_i + P_i c_i \quad P_i = a_i \hat{} b_i \quad G_i = a_i b_i \quad (7)$$

Where S_i, c_{i+1} are the sum and carry functions. P_i, G_i are the carry propagator and generator functions.

A 4-bit CLA adder is shown by using ULG with MRL logic [1]. But there are some problems in this adder there are c_1 is the extra carry applied to the circuit. Some unnecessary logic is used to implement c_{out} which causes the delay to produce the last carry. To overcome this, some logic architecture has to be modified.

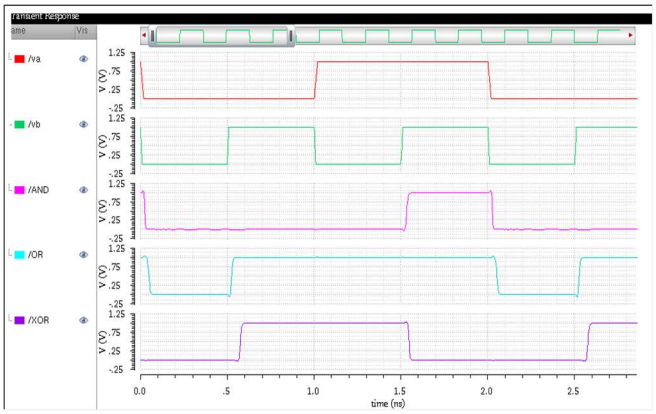


Fig. 8. MRLUG output

IV. AREA EFFICIENT CLA ADDER

In C_2 case, $C_{in}P_0$ pre-calculated in C_1 so using another 2-input AND gate here instead of 3-input AND gate. In C_3 case, $C_{in}P_0P_1$& G_0P_1 is pre-calculated in C_2 so using it another 2-i/p AND gate here instead of 4-input AND gate. In C_4 case, $C_{in}P_0P_1P_2$, $G_0P_1P_2$, G_1P_2 is pre-calculated in C_2 so using it another 2-i/p AND gate here. By replacing certain logics without changing the logics we are able to reduce Area

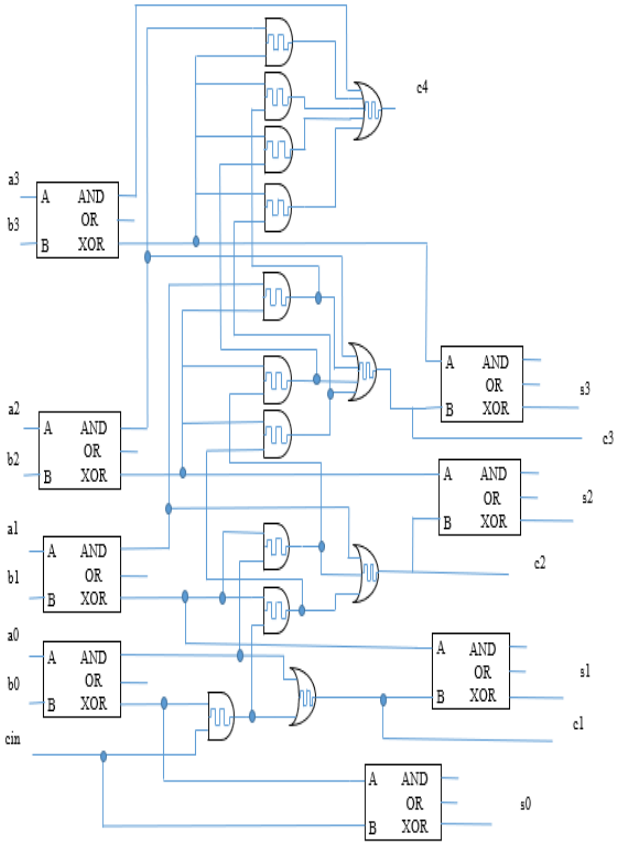


Fig. 9. A 4-bit CLA Adder

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overhead, unnecessary delay also reduced. If the memristor logic gate contains more inputs, then the count of memristors also increase. By reducing the inputs to the logic gate so this can help to save area, decrease the unnecessary delay.

V. RESULTS AND DISCUSSION

Area efficient CLA Adder has been implemented by using Cadence Virtuoso. VTEAM model is used here. The parameters are given in [7]. The number of memristors reduced from 75 to 66.

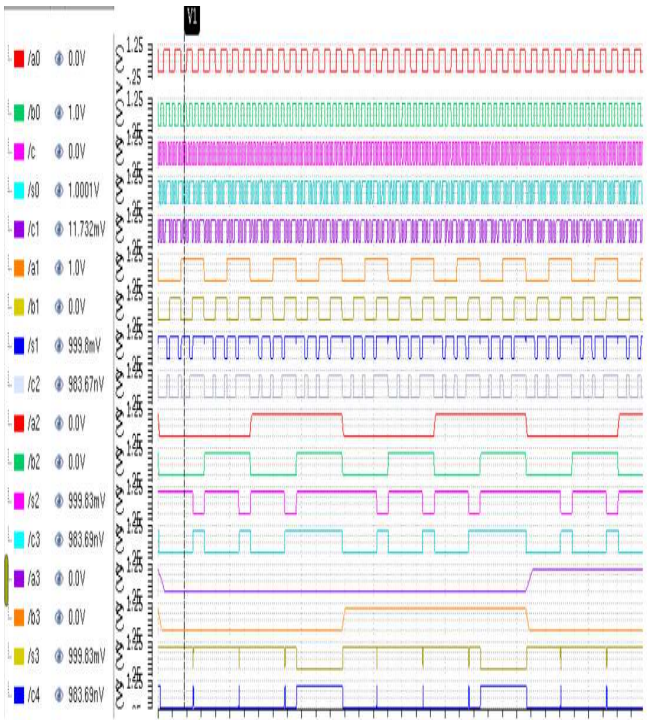


Fig. 10. A 4-bit CLA Adder output

The memristors count reduced from 75 to 66. The size of the CMOS used in ULG is 90nm, and the thickness of the memristor is 3nm.

TABLE I COMPARISON BETWEEN MODIFIED AND PROPOSED WORK

Parameter	Modified Work	Proposed Work
Logic Count	Memristors 66 Transistors 16	Memristors 75 Transistors 16
Technology (in nm)	Memristors 3 Transistors 90	Memristors 3 Transistors 180
Area (in mm^2)	1.72	2.45
Delay of carry out (in ns)	27.78	32.56

VI. CONCLUSION

In summary, we designed an area-efficient 4-bit CLA adder. The propagation delay has been decreased by decreasing the number of memristor which decreases the consumption of power. This design can be used in high bit adders, subtractors, multipliers.

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